A Tutorial on K-Means Clustering: Understanding Cluster Formation and the Impact of Choosing K

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Module: Machine Learning
Dataset: Mall Customer Dataset

1. Introduction:

Often, real-world data does not come with labels, so it is hard to use supervised learning methods right away. In these situations, unsupervised learning helps us find patterns and structure in the data. Clustering is one of the most common unsupervised techniques.

This tutorial will cover K-Means clustering, a straightforward but effective way to group similar data points. We will explain how K-Means works using the Mall Customer dataset, focusing on how clusters are created and why choosing the right number of clusters (K) matters.

Clustering is used in many fields, like customer segmentation and pattern recognition, where it is important to find hidden patterns in data (Han, Kamber and Pei, 2012).

2. Position of K-Means in Machine Learning:

Machine learning methods are usually grouped into supervised, unsupervised, and semi-supervised learning.

- **Supervised learning** uses labeled data and is often used for classification and regression tasks.
- **Unsupervised learning** works with unlabeled data and looks for patterns in the dataset.

K-Means clustering is an **unsupervised learning** method and is one of the most popular clustering algorithms (Pedregosa et al., 2011). Clustering techniques like K-Means group similar data points together based on their features.

3. Overview of K-Means Clustering:

K-Means is an algorithm that partitions data into a predefined number of clusters, where each cluster is represented by a central point called a **centroid**.

The algorithm works by minimising the **within-cluster variance (inertia)**, which measures how close data points are to their assigned centroid (MacQueen, 1967).

The steps of the K-Means algorithm are as follows:

1. Choose the number of clusters (K)
2. Initialise K centroids randomly
3. Assign each data point to the nearest centroid
4. Recalculate the centroids based on assigned points
5. Repeat until convergence

K-Means typically uses Euclidean distance to measure similarity (Han, Kamber and Pei, 2012).

4. **Dataset Description:**

The dataset used in this tutorial is the **Mall Customer dataset**, which contains information about customers such as gender, age, annual income, and spending score.

For this tutorial, two features are selected:

- **Annual Income (k$)**
- **Spending Score (1–100)**

These features allow clear visualisation of clustering results.

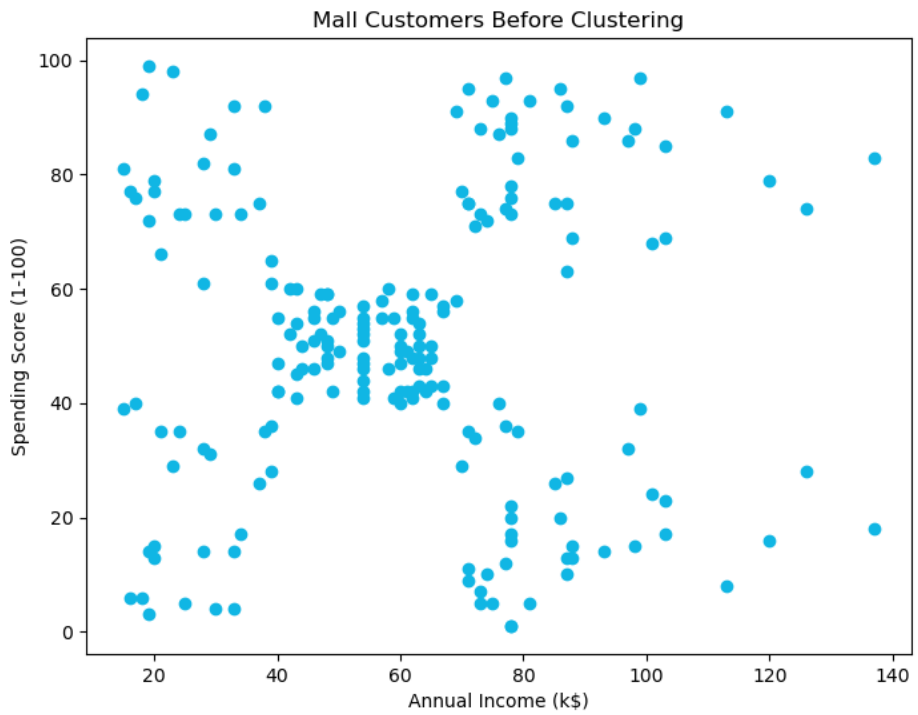
5. **Data Preparation:**

Before applying K-Means, the dataset is examined to ensure there are no missing values. The selected features are then standardised to ensure equal contribution to distance calculations.
Feature scaling is essential because K-Means is sensitive to differences in scale between variables (Pedregosa et al., 2011).

6. **Visualising the Data:**

The dataset is first visualised using a scatter plot (**Plot 1**), showing annual income versus spending score.

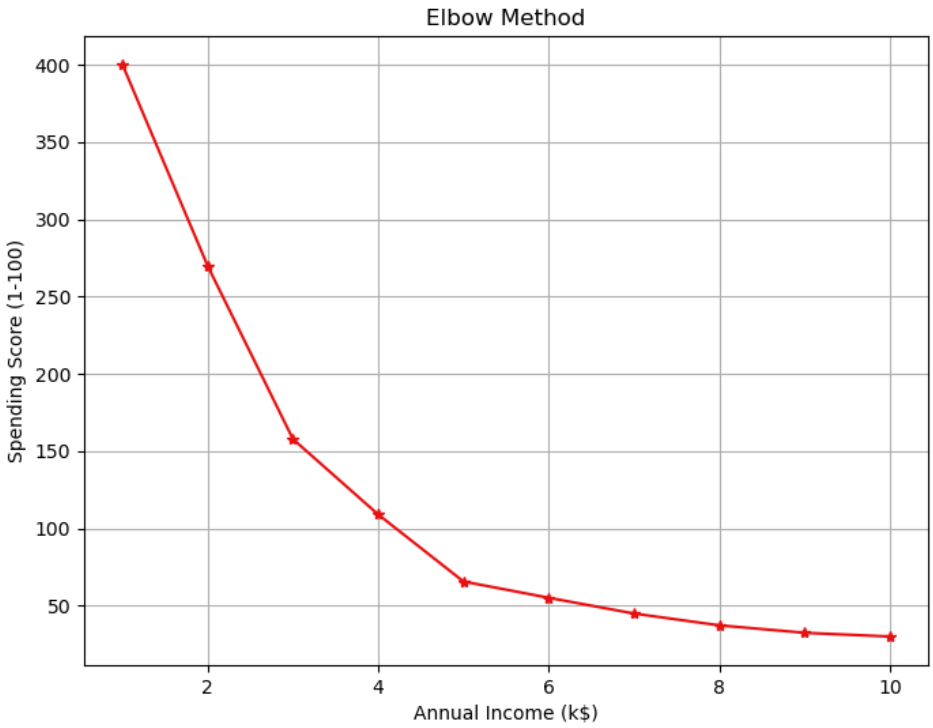
As shown in **Plot 1**, the data points are scattered without clearly defined groups. This demonstrates the need for clustering techniques to uncover hidden patterns in the dataset.



7. **Choosing the Number of Clusters (K):**

Selecting the correct number of clusters is a critical step in K-Means clustering.
The **elbow method** is used to determine an appropriate value of K. This method plots inertia against different values of K (**Plot 2**).

As seen in **Plot 2**, inertia decreases rapidly at first and then begins to level off. The “elbow” point suggests that **K = 5** is a suitable choice, balancing accuracy and simplicity.

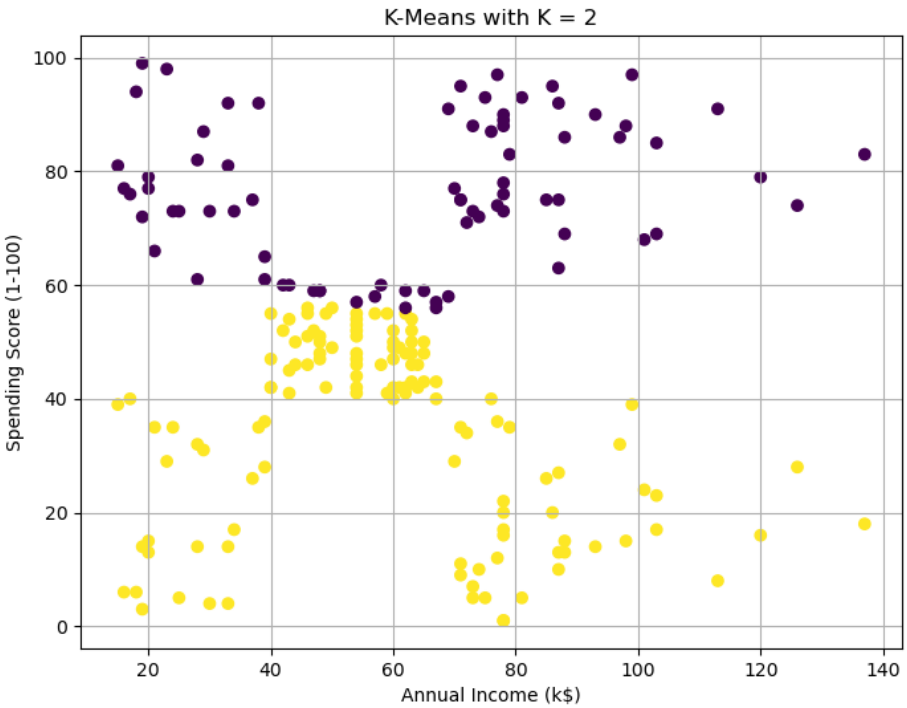


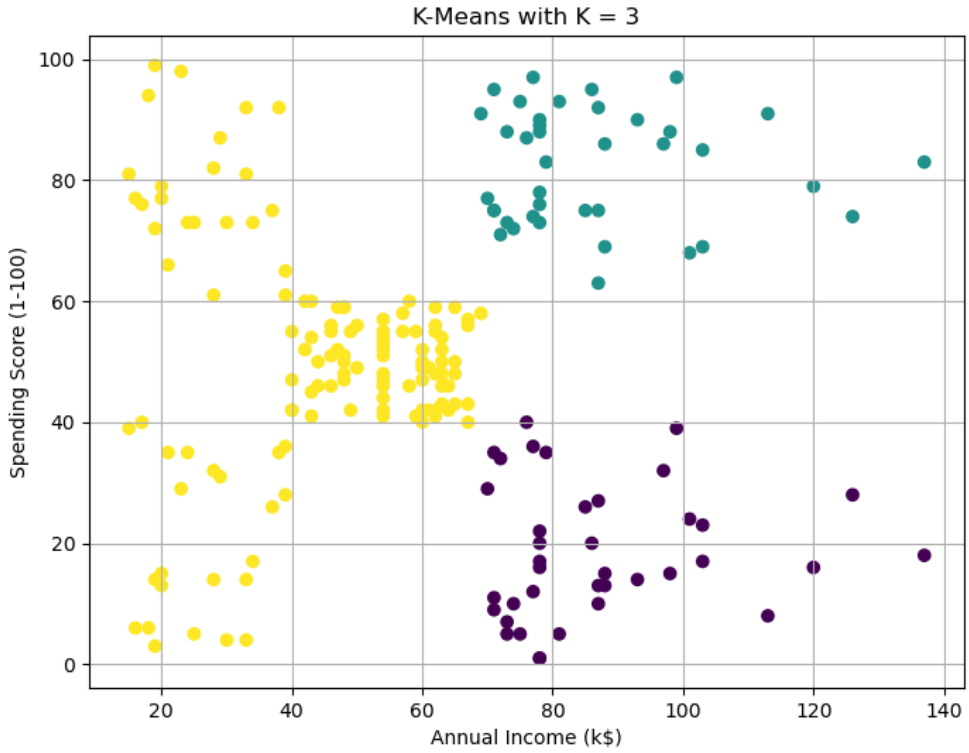
8. **Effect of Different Values of K:**

To better understand the impact of K, clustering results are compared for different values (**Plots 3 and 4**).

- In **Plot 3 (K = 2)**, the clustering is too broad and does not capture meaningful group differences.
- In **Plot 4 (K = 3)**, the grouping improves, but some clusters remain unclear.

These results demonstrate that using too few clusters can oversimplify the data and hide important patterns.

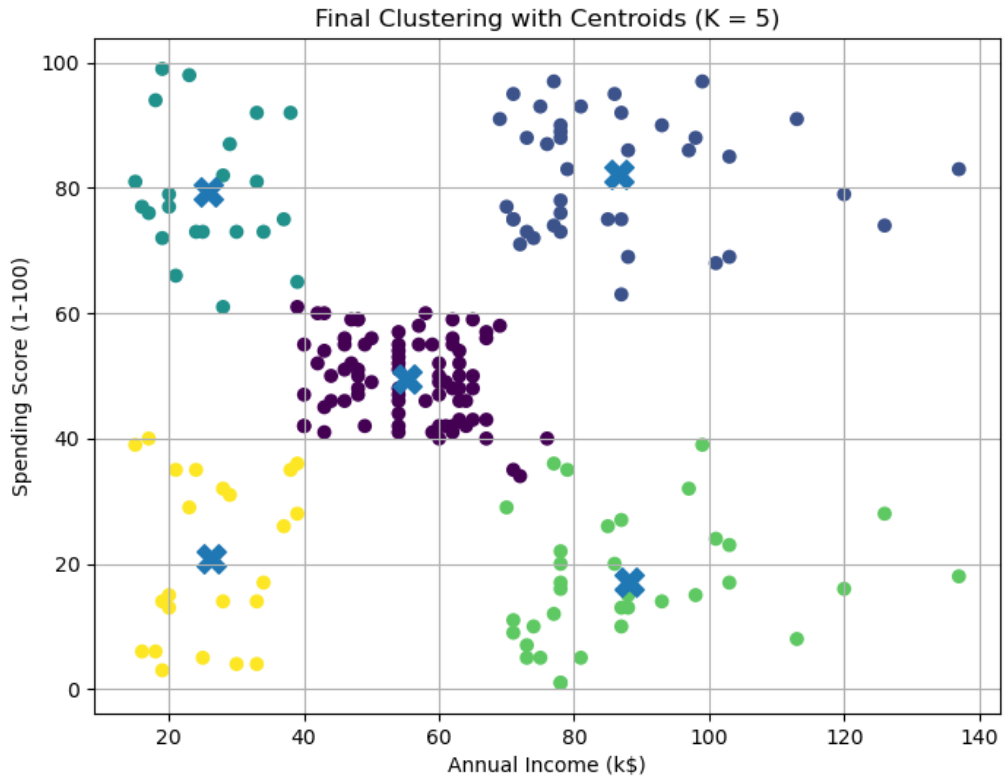




9. Final Clustering Results:

The final clustering is performed using **K = 5**, as suggested by the elbow method.

As shown in **Plot 5**, the clusters are clearly separated, and centroids are marked to represent the centre of each group. This provides a more meaningful segmentation of customers based on their income and spending behaviour.



10. Limitations of K-Means:

Despite its effectiveness, K-Means has several limitations:

- The value of K must be selected in advance
- Results depend on initial centroid placement
- It assumes clusters are spherical and evenly sized
- It is sensitive to outliers

These limitations should be considered when interpreting results (Han, Kamber and Pei, 2012).

11. Conclusion:

This tutorial showed how you can use K-Means clustering to group similar data points when you do not have labelled data. By looking at the results and trying different values for K, you can see that your choice of K has a big impact on how well the clustering works.

By using the elbow method and comparing clusters, we found that K = 5 works well for this dataset. The final groups give useful insights into how customers behave.
In summary, K-Means is a straightforward and useful tool for exploring data, but it is important to choose the right settings and interpret the results carefully.

12. References:

- MacQueen, J. (1967) *Some methods for classification and analysis of multivariate observations*. Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability.
- Pedregosa, F. et al. (2011) 'Scikit-learn: Machine learning in Python', *Journal of Machine Learning Research*, 12, pp. 2825–2830.
- Han, J., Kamber, M. and Pei, J. (2012) *Data Mining: Concepts and Techniques*. 3rd edn. Morgan Kaufmann.
- Scikit-learn (2024) *KMeans documentation*. Available at: https://scikit-learn.org/